Udemy Courses Analysis

Project Report

Exploratory analysis of 3,672 Udemy courses with Python, pandas and Seaborn

1. Introduction

This project explores a dataset of Udemy courses in four subjects. It asks which courses, subjects and levels attract the most students, whether price affects popularity, how free and paid courses compare, and which areas generate the most estimated revenue.

2. Objectives

- Clean the data and summarise its main numeric features.
- Identify the most popular courses by subscribers, reviews, lectures and duration.
- Test the relationship between price, subscribers, reviews and revenue.
- Compare subjects, levels and free vs paid courses.
- Estimate revenue per course and compare it across subjects and levels.
- Explore Python-related courses.

3. Tools and Technologies

Tool	Purpose
Python	Programming language
pandas / NumPy	Cleaning, filtering, grouping, aggregation
Matplotlib / Seaborn	Bar plots and scatter plots
Jupyter / Google Colab	Interactive development environment

4. Dataset

The dataset has **3,672 rows and 12 columns** after duplicates are removed, and no missing values. Columns: course_id, course_title, url, is_paid, price, num_subscribers, num_reviews, num_lectures, level, content_duration, published_timestamp and subject. The four subjects are Business Finance, Graphic Design, Musical Instruments and Web Development; the four levels are All Levels, Beginner, Intermediate and Expert.

Summary statistics

Statistic	Price ($)	Subscribers	Reviews	Lectures	Duration (h)
Mean	66.10	3,190.6	156.4	40.1	4.10

Statistic	Price ($)	Subscribers	Reviews	Lectures	Duration (h)
Std. dev.	61.04	9,488.1	936.2	50.4	6.06
Min	0	0	0	0	0
Median	45	912	18	25	2.0
Max	200	268,923	27,445	779	78.5

The median subscriber count (912) is far below the mean (3,191), so a small number of very popular courses dominate the totals.

5. Methodology

- **Load and clean:** read the CSV, apply `drop_duplicates()` and check `isnull().sum()`.
- **Rank:** find maximum subscribers, reviews, lectures and duration; list the top 10 by subscribers.
- **Correlation:** Pearson correlation of price with reviews, subscribers and revenue, plus scatter plots.
- **Group comparison:** count, mean and sum of subscribers by subject, level and paid/free status.
- **Revenue estimate:** new column `revenue = num_subscribers * price`, compared by subject and level.
- **Text search:** `str.contains('Python')` on course titles.

6. Findings

6.1 Free vs paid

Measure	Paid	Free
Number of courses	3,362	310
Total lectures	140,756	6,639
Average subscribers	2,430	11,534
Total subscribers	8,183,595	3,575,525

About 92% of courses are paid, but the average free course attracts almost five times as many subscribers as the average paid course.

6.2 Most popular courses

Measure	Course	Value
Subscribers	Learn HTML5 Programming From Scratch	268,923
Reviews	The Web Developer Bootcamp	27,445
Lectures	Back to School Web Development and Programming...	779
Duration	The Complete Figure Drawing Course HD	78.5 hours
Est. revenue	The Web Developer Bootcamp	$24,316,800

Seven of the top ten courses by subscribers are web development or programming courses; one is a free guitar course.

6.3 Subjects

Subject	Courses	Avg subscribers	Total subscribers	Total est. revenue
Web Development	1,200	6,650	7,980,572	$630.8M
Business Finance	1,195	1,564	1,868,711	$123.7M
Graphic Design	603	1,763	1,063,148	$77.0M
Musical Instruments	680	1,245	846,689	$53.4M

Web Development leads on every measure, with roughly four times the average subscribers of any other subject. Musical Instruments is the smallest in subscribers and revenue.

6.4 Levels

Level	Courses	Avg subscribers	Total subscribers	Total est. revenue
All Levels	1,929	3,585	6,915,076	$610.0M
Beginner	1,270	3,190	4,051,843	$219.7M
Intermediate	421	1,762	742,005	$49.7M
Expert	58	865	50,196	$5.5M

Courses open to all levels and beginners draw most students. Demand falls sharply with difficulty, and Expert courses are few and attract the fewest subscribers.

6.5 Correlations

Pair	Correlation	Interpretation
Price vs reviews	0.11	Weak
Price vs subscribers	0.05	Almost none
Price vs revenue	0.35	Weak to moderate
Subscribers vs revenue	0.56	Moderate

Price has little effect on how many students enrol. Revenue depends more on subscriber count than on price, though it is partly built from both by definition.

6.6 Python courses

29 course titles contain 'Python'. The most subscribed is *Web Programming with Python* (35,267), followed by *Coding for Entrepreneurs: Learn Python, Django...* (23,412) and *Learn Python and Django: Payment Processing* (17,714).

7. Issues and Recommended Improvements

#	Issue	Recommended fix
1	The paid vs free bar chart uses hard-coded counts [310, 3362] against labels ['Paid Course', 'Free Course'], so the paid bar shows 310. The correct counts are 3,362 paid and 310 free.	Compute counts with `data['is_paid'].value_counts()` instead of typing them.

#	Issue	Recommended fix
2	The 'total reviews by subject' cell sums `num_subscribers`, so the output repeats the subscriber totals.	Sum `num_reviews`.
3	Revenue is estimated as subscribers x list price. It ignores discounts, refunds and the platform's share, so it overstates real earnings.	Label it 'estimated gross revenue' and state the assumption.
4	Correlations of 0.35 and 0.56 are called 'negligible' and 'mediocre'.	Describe them as weak-to-moderate; consider Spearman correlation because the data is heavily skewed.
5	Several answers use a hard-coded value to find the top course (e.g. `== 268923`).	Use `idxmax()` so results update with the data.
6	Seaborn is used before it is imported, and counts differ between cells (3,362 vs 3,368 paid courses), which suggests cells were run out of order.	Move imports to the top and use Restart and Run All.
7	`plt.figure(figsize=(100,10))` is called after the plot, creating an empty oversized figure.	Call it before the plot with a sensible size such as (10, 5).
8	Subject, level and paid/free comparisons are written as long repeated print blocks. Python search is case-sensitive.	Use `groupby().agg()` and `str.contains('python', case=False)`.
9	Data is read from a Colab Drive path.	Use a relative path.

8. Conclusion

Web Development dominates the Udemy dataset: it has the most courses, subscribers and estimated revenue. Courses suitable for all levels and for beginners are the most popular, while Expert courses attract few students. Price has almost no relationship with subscriber count, and free courses attract far more students per course than paid ones. The notebook would be more reliable with the chart label and review-sum fixes above, careful wording about correlation strength, and revenue clearly labelled as an estimate.

9. Future Work

- Analyse `published_timestamp` to study how course supply and popularity change over time.
- Add a correlation heatmap and histograms of the skewed variables.
- Compare review rate (reviews per subscriber) as a measure of engagement.
- Study how duration and lecture count relate to subscribers.
- Analyse title keywords (e.g. 'beginner', 'complete', 'bootcamp') against popularity.
- Build a simple model to predict subscribers from price, level, subject and duration.